



Powers of Two

Problem 686

$2^7 = 128$ is the first power of two whose leading digits are "12".

The next power of two whose leading digits are "12" is 2^{80} .

Define $p(L, n)$ to be the n th-smallest value of j such that the base 10 representation of 2^j begins with the digits of L .

So $p(12, 1) = 7$ and $p(12, 2) = 80$.

You are also given that $p(123, 45) = 12710$.

Find $p(123, 678910)$.